Name_____ Date_____ Period_____

LESSON 6-8 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Using Properties to Rewrite Expressions

Maryland Standard 6.EE.A.3 · Learning target – I can use the distributive property and like terms to rewrite an expression in simplest form.

REMEMBER

- **Distributive property:** multiply the outside number by **each** term inside: $4(x + 5) = 4x + 20$.
- **Commutative:** change the order. **Associative:** change the grouping.
- Simplest form: distribute, put like terms together, then combine them.

WORKED EXAMPLE

Write $3(k + 4) + 2k$ in simplest form.

Step 1 Distribute: $3k + 12 + 2k$

Step 2 Reorder (commutative): $3k + 2k + 12$

Step 3 Combine like terms: $5k + 12$

Answer $\mathbf{5k + 12}$

WORD BANK

distribute – multiply the outside number by every term inside

like terms – terms with the same variable, like 4c and 5c

simplest form – no like terms left to combine

PRACTICE

show your work · label units

1. Use the distributive property to rewrite each expression.

Step 1 Multiply the outside number by the first term.

Step 2 Multiply it by the second term too. Keep the + or − sign.

a. $6(n + 2) =$ _____ **b.** $3(4x + 1) =$ _____ **c.** $5(2 + y) =$ _____

d. $7(a - 3) =$ _____ **e.** $2(3b + 5) =$ _____ **f.** $10(m - 1) =$ _____

2. Use the distributive property to find each product. Show the sum you used.

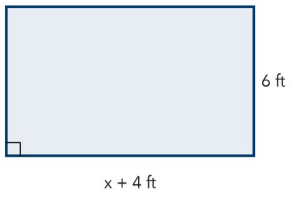
a. $6 \times 34 =$ _____ **b.** $8 \times 52 =$ _____

3. Write each expression in simplest form.

a. $4(c + 3) + 5c =$ _____ **b.** $2(x + 6) + x + 1 =$ _____

c. $3(m + 2) + 4(m + 1) =$ _____ **d.** $8 + 5(t + 2) =$ _____

4. Garden. A garden is 6 feet **wide** and (x + 4) feet **long**. Its area is 6(x + 4) square feet. Write the area **without parentheses**. Then find the area when x = 5.



Step 1 Multiply 6 by x and 6 by 4.

Step 2 Put 5 in for x and find the value.

Show your work.

Answer_____

5. Find the Error. Ana simplified 2(3x + 4) + x and got 7x + 4. Explain her mistake. Then simplify correctly.

Frame *Ana forgot to multiply ___ by 2. The correct answer is ___.*

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Show someone at home how 3(10 + 2) = 30 + 6 helps you find 3 × 12 in your head.

Name_____ Date_____ Period_____

LESSON 6-8 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Expanding and Factoring Expressions

Maryland Standard 6.EE.A.3 · Learning target – I can expand an expression and factor one using the greatest common factor.

REMEMBER

- **Expand:** distribute to remove parentheses: $5(2x + 3) = 10x + 15$.
- **Factor:** put the GCF outside the parentheses; divide each term by it.
- Check a factored form by expanding it.

WORKED EXAMPLE

Factor $24y + 36$.

Step 1 The GCF of 24 and 36 is 12. Divide: $24y \div 12 = 2y$ and $36 \div 12 = 3$

Step 2 Write $12(2y + 3)$. Check: $12 \times 2y + 12 \times 3 = 24y + 36$

Answer **$12(2y + 3)$**

WORD BANK

expand – multiply to remove the parentheses

factor – write as a number times (___ + ___)

GCF – the biggest number that divides both numbers

PRACTICE

show your work · label units

1. Expand each expression.

a. $3(5x + 2) =$ _____ **b.** $8(y + 7) =$ _____ **c.** $4(3 + 2n) =$ _____

2. Factor each expression using the GCF.

Step 1 Find the GCF of the two numbers.

Step 2 Divide each term by the GCF.

Step 3 Write GCF(___ + ___) and check by expanding.

a. $6x + 15 =$ _____ **b.** $20y + 32 =$ _____ **c.** $14a + 49 =$ _____

d. $36 + 27k =$ _____ **e.** $40m + 60 =$ _____ **f.** $11b + 33 =$ _____

3. **Rectangle.** A rectangle has an **area** of $8x + 56$ square units. Its **width** is 8 units. Write an expression for its **length**.

Show your work.

Length_____

4. **Stretch.** Write two different factored forms of $12x + 24$. Then write the form that uses the GCF.

5. **Reason.** Leon factored $30n + 45$ as $5(6n + 9)$. Did he use the GCF?

Frame *6n and 9 share the factor ___. So Leon ___ use the GCF. The GCF form is ___.*

□ Yes □ No

TEST PRACTICE

choose the best answer

6. Which is $16x + 40$ factored using the GCF?

Step 1 Find the GCF of 16 and 40.

Step 2 Pick the choice with the GCF outside the parentheses.

A $2(8x + 20)$ **B** $8(2x + 5)$ **C** $4(4x + 10)$ **D** $8(2x + 40)$

7. Which expression is equivalent to $6(3a + 2)$?

A $18a + 2$ **B** $9a + 8$ **C** $18a + 12$ **D** $3a + 12$

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Show someone at home how to factor $6x + 9$ as $3(2x + 3)$. Then expand it to check.